

THE PHASE I PROPOSAL SUBMISSION DEADLINE FOR THE FOLLOWING TOPICS IS MAY 10, 2023 AT 2:00PM EASTERN DAYLIGHT TIME (EDT).

THE DEADLINE TO SUBMIT QUESTIONS RELATED TO THE FOLLOWING TOPICS IS MARCH 28, 2023.

CBA 01 - Detection: Bioinformatics Capabilities for the Field and Beyond

Background: Recent advancements in biological sequencing technology have shown the ability to address emerging, new, minimally characterized, or engineered threats and supplement currently fielded detection capabilities using near real-time threat identification capabilities. While the development of such technologies has allowed the Warfighter to leverage commercial-off-the-shelf (COTS) advancements to perform sequencing in the field, additional improvements to data analysis, visualization and reporting are needed to ensure the user is given clear, concise results that are actionable. Current data outputs are highly technical and are not easily interpreted by general purpose forces. Additionally, general purpose forces need information to quickly and accurately discern a potential threat or hazard versus something that is benign to support the mission.

Objective: This topic seeks novel bioinformatics, data analysis and reporting solutions for all existing sequencing technologies, but will pay special focus to portable sequencing platforms that can be used out in the field. Sequencing analysis solutions should have the ability to work on commodity hardware such as a tablet or laptop without dependency or reliance to high performance computing solutions. Heavy focus will be on data interpretation, visualization and reporting to aid decision making of complex biological data in a field environment. The capability should address the following (preference will be given to proposals that address all areas):

1. Provides rapid, real-time data analysis focused on discerning whether or not threat exists from sequencing information and provide clear reporting for in-field interpretation with the ability to provide detailed, technical sequencing reports for additional analysis as needed.
2. Graphical User Interface (GUI) and data interpretation need to be intuitive and showcase results.
3. Development of new or modification of existing bioinformatic solutions shall be applicable to accurate microbial detection, identification, and threat prediction based on sequencing data ingested and analyzed.
4. Data analysis should be flexible and adaptable with the ability to run in lean computing conditions such as working on a laptop, tablet, or other portable hardware, but also have the ability to scale to high performance capabilities for advanced analytics.

Applicants are encouraged to develop research collaborations with other Government agencies, academic institutions, and private entities. In addition, applicants can leverage resources within the Department of Defense research laboratories.

Impact: This topic will advance biological threat detection technologies to support joint force readiness against potential biological threats. This capability is crucial to allow an accurate, rapid and concise threat detection and identification following a biological warfare agent (BWA) exposure event (deliberate or natural), avoid unnecessary personnel exposure, and support early warning goals addressing emerging biological threats.

References:

J.A. Carriço, M. Rossi, J. Moran-Gilad, G. Van Domselaar, M. Ramirez, A primer on microbial bioinformatics for nonbioinformaticians, *Clinical Microbiology and Infection*, Volume 24, Issue 4, 2018, Pages 342-349, ISSN 1198-743X, <https://doi.org/10.1016/j.cmi.2017.12.015>.

Gamaarachchi, H., Lam, C.W., Jayatilaka, G. *et al.* GPU accelerated adaptive banded event alignment for rapid comparative nanopore signal analysis. *BMC Bioinformatics* **21**, 343 (2020). <https://doi.org/10.1186/s12859-020-03697-x>

Haowen Zhang, Haoran Li, Chirag Jain, Haoyu Cheng, Kin Fai Au, Heng Li, Srinivas Aluru, Real-time mapping of nanopore raw signals, *Bioinformatics*, Volume 37, Issue Supplement_1, July 2021, Pages i477–i483, <https://doi.org/10.1093/bioinformatics/btab264>

Rory Munro, Roberto Santos, Alexander Payne, Teri Forey, Solomon Osei, Nadine Holmes, Matthew Loose, minoTour, real-time monitoring and analysis for nanopore sequencers, *Bioinformatics*, Volume 38, Issue 4, 15 February 2022, Pages 1133–1135, <https://doi.org/10.1093/bioinformatics/btab780>

Wang Y, Zhao Y, Bollas A, Wang Y, Au KF. Nanopore sequencing technology, bioinformatics and applications. *Nat Biotechnol.* 2021 Nov;39(11):1348-1365. doi: 10.1038/s41587-021-01108-x. Epub 2021 Nov 8. PMID: 34750572; PMCID: PMC8988251.

CBA 02 - Diagnostics: Analysis of Agent Exposure Biomarkers in the Ocular System

Background: Warfighters are often deployed to austere environments under potential exposure to adversary or naturally occurring chemical and biological warfare agents (CBWA). Rapid diagnostics tools against CBWAs are crucial for battlefield readiness and countermeasure deployment (i.e. medical and strategic). The majority of diagnostics platforms utilize traditional clinical samples (e.g. blood, urine) to test for host- or pathogen-based biomarkers. However, advances in proteomic and imaging analyses of the ocular system revealed the potential for a new source of biomarkers for systemic and neurodegenerative diseases¹. The Defense Threat Reduction Agency is exploring the feasibility to leverage the ocular system for the development of novel sampling and diagnostic tools against CBWAs.

The ocular system provides a unique source of minimally- or non-invasive sampling matrices for medical diagnostics including tears and the retina. Proteomic analysis of tear samples identified candidate biomarkers for Alzheimer's disease and diabetic retinopathy^{2,3,4}. In addition, inflammatory cytokines and chemokines are elevated in tears from ocular diseases patients¹, suggesting the potential adaptability of established immuno-based biomarkers to tear samples. Recent studies also revealed differentially expressed miRNAs in tears from Alzheimer's disease patients⁵. Altogether, these studies highlight the richness of analytes present in tears and the potential for biomarker discovery at both the protein and nucleic acid level.

Ocular imaging is an emerging tool for diagnostics of neurodegenerative diseases⁶. In particular, retinal imaging studies found measurable structural and functional changes associated with neurodegeneration and neuroinflammation in these patients. The potential use of retinal imaging for infectious diseases and exposure to chemical agents are yet to be determined. However, it is plausible that CBWA targeting the nervous system or inducers of an inflammatory response could be detected using ocular imaging.

Objective: This topic seeks innovative approaches for minimally or non-invasive ocular sampling technologies for the diagnosis of CBWA exposures. Proof-of-concept studies for technologies or assays under development are welcomed. Successful efforts should address any or all of the following (preference will be given to proposals that address more areas):

1. Identify relevant biomarkers in minimally or non-invasive ocular samples (e.g. tears, eye imaging, etc.).
2. Develop minimally or non-invasive methods for sample collection or eye imaging in the field.
3. Diagnostics tool should be adaptable to agnostic or multiplexed analyses against CBWAs.
4. Wearables or other technologies (e.g. scleral lenses) for continuous monitoring of ocular biomarkers.

Applicants are encouraged to develop research collaborations with other Government agencies, academic institutions, and private entities. In addition, applicants can leverage resources within the Department of Defense research laboratories.

Impact: This topic will advance medical diagnostics technologies to support warfighters readiness against potential CBWA threats. The envisioned tool will provide a novel diagnostic tool independent of traditional sample matrices (i.e. blood and urine) and expand our knowledge of biomarkers present in the ocular system. These capabilities are crucial to allow rapid diagnostics following a CBWA exposure event (deliberate or natural), avoid unnecessary treatment, and prioritize appropriate medical care.

References:

- ¹ Tamhane, M., Cabrera-Ghayouri, Abelian, G., and Viswanath, V. (2019) Review of Biomarkers in Ocular Matrices: Challenges and Opportunities. *Pharmaceutical Research* 36(3): 40.
- ² Chalkias, I., Tegos, T., Topouzis, F., and Tsolaki, M. (2021) Ocular Biomarkers and Their Role in the Early Diagnosis of Neurocognitive Disorders. *European Journal of Ophthalmology* 31:6, 2808-2817.
- ³ Csosz, E., Boross, P., Csutak, A., *et al.* (2012). Quantitative Analysis of Proteins in the Tear Fluid of Patients with Diabetic Retinopathy. *Journal of Proteomics* 75(7): 2196-2204.
- ⁴ Ponzini, E., Santambrogio, C., de Palma, A., *et al.* (2022). Mass Spectrometry-based Tear Proteomics for Noninvasive Biomarker Discovery. *Mass Spectrometry Reviews* 41(5): 842-860.
- ⁵ Kenny, A., Jimenez-Mateos, E.M., Zea-Sevilla, M.A., *et al.* (2019). Proteins and microRNAs are Differentially Expressed in Tear Fluid from Patients with Alzheimer's Disease. *Science Reports* 9: 15,437.
- ⁶ Christinaki, E., Kulenovic, H., Hadoux, X., *et al.* (2022). Retinal Imaging Biomarkers of Neurodegenerative Diseases. *Clinical Experimental Optometry* 105(2): 194-204.

CBS 01 - Emerging Threats: Discovery and Development of Broad-Spectrum, Cross Toxidromic Chemical Medical Countermeasures

Background: Toxidromes are constellations of signs, symptoms and/or characteristics associated with exposures to a specific class of toxic chemicals, such as anti-cholinergic, cholinergic, opioid, convulsant, anticoagulant, stress-response/sympathomimetic, and knockdown/asphyxiant agents (<https://chemm.hhs.gov/toxicsyndromes.htm#sec3>). While the specific mechanism(s) of toxicity of the agents that cause these various toxidromes are diverse, there can be overlaps in how they disrupt homeostasis and physiology. These shared mechanistic elements of the pathological process raise the possibility of formulating medical countermeasures that act on the host side to restore homeostasis independent of the specific toxidrome, and thus provide therapeutic benefit against a broad-spectrum of agents.

For chemical warfare agents (CWAs) and other chemical emerging threat agents that are rapidly lethal, toxidrome recognition and treatment with antidotes, emergency respiratory support, and/or decontamination may be more effective than a time-consuming, agent-specific, definitive diagnosis. Medical countermeasures (MCMs) that target the toxidrome class are often sufficient for immediate treatment. Therefore, identifying medical countermeasures that can be used across multiple toxidromes may enable first responders, including combat medics, to save more lives. Over the years, some have suggested similar approaches, for example using antioxidants to treat lung, dermal and/or ocular injury of a variety CWAs and toxic industrial chemicals (cite McElroy and Cowan).

Objective: Identify, down-select, and develop candidate broad-spectrum medical countermeasures (prophylactic or therapeutic) for emerging chemical threats. Candidates should have potential to act across multiple chemical toxidromes (e.g. Opioid, Cholinergic, Anticholinergic, Convulsant, Sedative, etc., <https://chemm.hhs.gov/toxicsyndromes.htm#sec3>).

Proposers are encouraged to consider applying Artificial Intelligence and Machine Learning (AI/ML) approaches to optimize predictions of agent: host target and MCM: therapeutic target interactions, since target sites that are highly conserved across toxidromes are more likely to demonstrate broad-spectrum cross-toxidromic efficacy. Additionally, it is expected that use of such AI/ML platforms will facilitate down selection of potential targets and expedite therapeutic design for new and emerging chemical threats.

1. Where applicable and to the extent feasible, repurposing FDA-approved drugs as MCMs to prevent or treat intoxication across multiple toxidromes is preferred.
2. Performers are encouraged to explore the utility of candidates known to work against one toxidrome as MCMs for additional toxidromes.
3. Efforts that include innovative AI/ML platforms to optimize predictions of therapeutic: target interactions are encouraged. If a target site is highly conserved across toxidromes / sub-species, then the candidate would likely show efficacy as a broad-spectrum countermeasure. This would also help expedite therapeutic design for new and emerging bio-threats.

4. MCMs that are effective across three or more toxidromes or syndromes will be given priority.
5. MCMs that are efficacious both peripherally and across the blood brain barrier (when relevant) will be given priority.
6. Proposers are encouraged to consider and address the potential for logistically supporting and employing the candidate countermeasure in operational settings. This includes storage temperature (ideally room temperature stable, 10°C to 35°C), stability in solution (ideally at least 90 days), and route of administration (ideally intramuscular; requirement for intravenous injection is strongly disfavored).

Impact: The results of proposed research efforts will ultimately provide the Warfighter and the Chemical and Biological Defense Program with a more agile and robust discovery and development pipeline for chemical medical countermeasures. Additionally, it will provide improved protection against and treatment for the adverse effects of the Cholinergic and/or Convulsant Toxidromes and the agents that cause them. Furthermore, this research will provide risk reduction to the current candidates in the pipeline for both the early and advanced developers.

Successful accomplishment of this effort will positively impact Warfighter readiness by unburdening medical logistics to fewer products able to address more requirements, and by streamlining time to treatment with less need for definitive diagnostics after a chemical exposure.

References:

1. McElroy C and Day BJ. 2016. Antioxidants as Potential Medical Countermeasures for Chemical Warfare Agents and Toxic Industrial Chemicals. *Biochem Pharmacol.* 2016 Jan 15; 100: 1–11. PMID: 26476351
2. Cowan FM, Broomfield CA, Stojiljkovic MP, Smith WJ. 2004. A Review of Multi-Threat Medical Countermeasures against Chemical Warfare and Terrorism. *Mil Med.* 2004 Nov; 169(11): 850–855. PMID: 15605928

CBS 02 - Discovery and Development of Novel, Non-Oxime Based Therapeutics to Treat Symptoms of Exposure to Organophosphorus Chemical Warfare Nerve Agents

Background: Organophosphorus (OP) chemical warfare agents (CWA) act by inhibition of acetylcholinesterase (AChE) resulting in reduced hydrolysis of the neurotransmitter, acetylcholine. Accumulation of acetylcholine causes overstimulation of the acetylcholine receptor. Physiological responses comprise hypotension, rhinorrhea, wheezing, and diarrhea among other symptoms, with severe exposures leading to death. OP intoxication poses a significant threat to military forces. Current medical countermeasures constitute utilize pralidoxime (2-PAM), a charged oxime reactivator of OP-inhibited AChE; atropine, an antimuscarinic agent that blocks the acetylcholine receptor; and diazepam to treat seizures resulting from excess acetylcholine in the central nervous system (CNS) by enhancing the inhibitory effects of gamma-aminobutyric acid in the brain.

A majority of research within this arena targets reactivation of OP-inhibited AChE using oxime-based compounds. Few groups have explored AChE reactivators containing different alternative functional moieties for reactivation, or pursued molecules with a mechanism of action other than the accepted oxime-based mechanism of action. Innovative approaches to ameliorate the biological effects of OP agent intoxication are desired, such as:

- Reactivation of AChE inhibited by a broad range of OP agents
- Alternate (non-AChE) methods to cleave excess acetylcholine
- Methods to facilitate reactivation of AChE inhibited by a broad range of OP agents
- Methods/compounds that to promote generation biosynthesis of new AChE

Objective: This topic seeks new and novel medical countermeasures against OP whose function is mechanistically distinct from the prevailing approaches that use oxime chemistry to reactivate AChE. Successful novel chemistries should match or exceed the performance of oxime countermeasures in efficacy for AChE reactivation, in the spectrum of OPs for which they are effective, and in their ability to treat the central nervous system effects of OP exposure. Research plans should span from inception (whether origination of novel candidates or identification of existing ones) through Phase I clinical studies. Phase I proposals should detail plans to move candidate compounds through TRL 6c (<https://www.medicalcountermeasures.gov/federal-initiatives/guidance/integrated-trls.aspx>).

Minimally, PIs should explain selection process for down selection of successful candidates and outline plans for in vitro assays to measure potency, structure-activity relationship (SAR) studies, Absorption/Distribution/Metabolism/Excretion (ADME) studies, in vivo work, chemistry and manufacturing, IND submission, and phase 1 clinical studies. The Government prefers to test lead compounds against live CW nerve agent such as sarin, soman, and/or VX rather than against surrogates or OP pesticides. If the performer is not able to work with live agent, the PI is strongly advised to team with a laboratory that is able conduct live agent testing. The goal is to advance lead compounds through Phase I clinical studies to support transition to Advanced Development. Candidates may be non-oxime reactivators of nerve agent inhibited AChE. Novel small molecules, peptides, peptidomimetics, or proteins will also be considered as potential therapeutics to treat nerve agent exposure. Candidates may also be drugs already approved by

the FDA or other health agencies which require additional testing to expand the existing indication to include treatment of OP exposure.

Note - This project is not interested in pursuit of protein or enzymes to stoichiometrically or catalytically hydrolyze nerve agent to remove from circulation, acetylcholine receptor inhibitors, AChE inhibitors, anticonvulsants, anti-inflammatories, gamma-aminobutyric acid or glutamate receptor modulators.

1. Develop lead candidate for proposed indication that performs better than the current standard of care (ie. more efficacious and/or more broad spectrum than 2-PAM for AChE reactivation).
2. Lead candidate should have physical properties appropriate to successful drug development.
3. Lead candidate should have an ADME profile that supports continuing development of that candidate.
4. Lead candidate should show efficacy in vivo for proposed indication (ie. AChE reactivation).
5. Lead candidate demonstrates an acceptable in vivo safety profile.
6. Lead candidate can be produced at the pilot scale in GMP-compliant lots.
7. Performer will ultimately prepare and submit IND
8. Performer will ultimately conduct Phase I clinical trials

Impact: Successful accomplishment of these objectives will expand the repertoire of available countermeasures for the treatment of OP CWA exposure. Key performance characteristics including efficacy against a broad spectrum of CWAs and ability to treat CNS symptoms of OP exposure will enhance the performance and recovery of Warfighters operationally exposed to CWAs.

References:

1. Gorecki L, Soukup O, Korabecny J. 2022. Countermeasures in organophosphorus intoxication: pitfalls and prospects. *Trends Pharmacol Sci.* 2022 Jul;43(7):593-606. PMID: 35643835
2. Worek F, Thiermann H, Wille T. 2020. Organophosphorus compounds and oximes: a critical review. *Arch Toxicol.* 2020 Jul;94(7):2275-2292. PMID: 32506210